



# What EKS Is

EKS is a managed Kubernetes control plane: AWS runs and scales the API server, scheduler, controllers, and etcd across availability zones, and you run the worker nodes and your workloads.

## IN THIS LECTURE

### 01 **Container orchestration**

The placement, healing, scaling and rollout problems Kubernetes solves.

### 02 **Desired state**

How controllers keep the cluster matching what you declared.

### 03 **Control plane vs data plane**

What decides and remembers, versus what runs your pods.

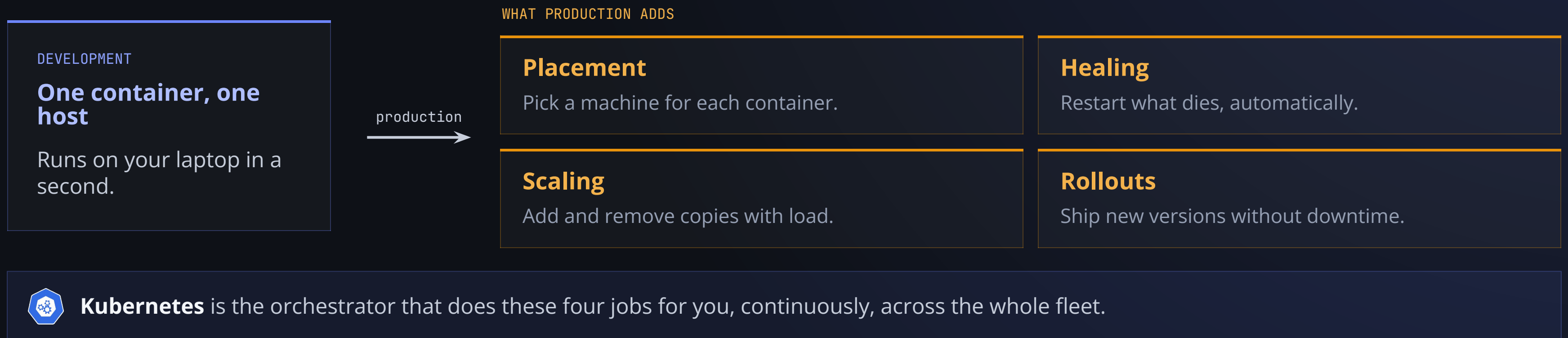
### 04 **The EKS split**

AWS operates the control plane, you operate the nodes.



# Placement, healing, scaling, rollouts

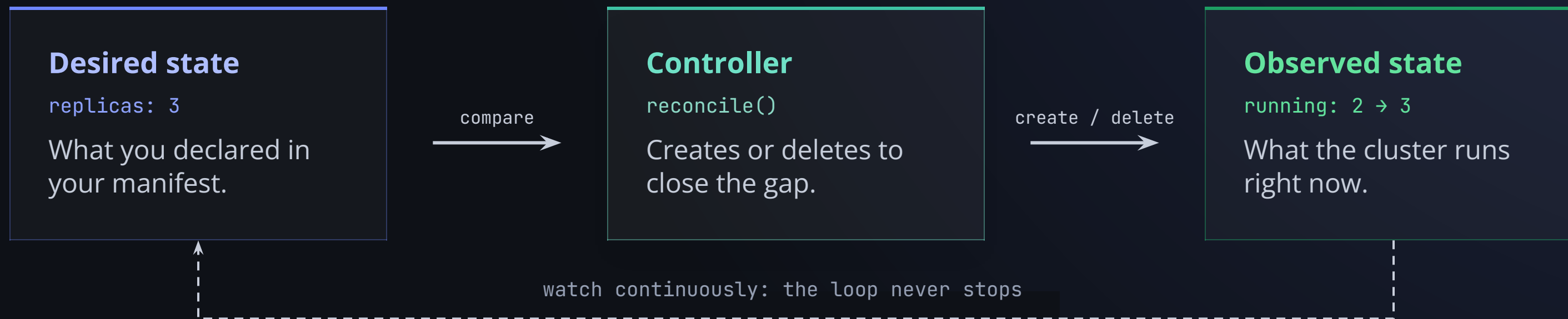
Running one container is trivial. Production runs hundreds across a fleet that fails and changes, and Kubernetes is the system that keeps them running as declared.



## DESIRED STATE

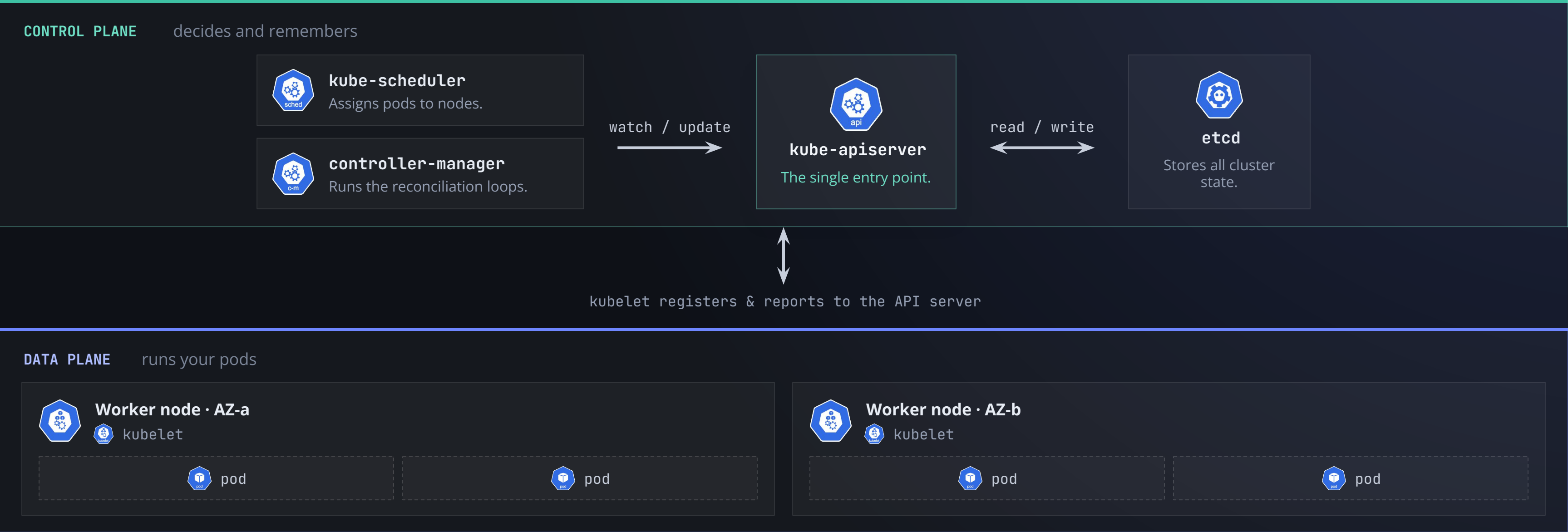
# You declare, controllers reconcile

Declare "three replicas." A controller compares that against what is running and, when they differ, creates or deletes pods until they agree. This loop runs forever, for every object.



# Control plane and data plane

The control plane holds the API server, scheduler, controllers, and the etcd datastore. The data plane is the worker nodes, each running a kubelet, where your pods actually execute.



# Running it yourself: etcd HA, patching, upgrades

Each of these keeps the cluster alive without improving your product, and etcd loss or a botched upgrade can take the whole cluster with it. None of it is visible to your customers.

## ! Highly available etcd

Replicate across zones, or a node loss takes the whole cluster.

## ! Backups and restore

etcd holds every object you declared; losing it loses everything.

## ! API server patching

Keep it secure, available, and scaled as the cluster grows.

## ! Version upgrades

Every control-plane component, in order, without downtime.

## ! Security hardening

The API server is the entry point to the entire cluster.

## ! Capacity and scaling

The control plane must grow with object and node count.

# EKS runs the control plane; you run the nodes

Create a cluster and AWS stands up and runs the API server, etcd, scheduler, and controllers. You bring the worker nodes and everything that runs on them.



## AWS operates

- ✓ The API server, scheduler, controllers, and etcd, on the infrastructure AWS owns.
- ✓ High availability across multiple availability zones.
- ✓ Patching, scaling, and a supported version-upgrade path.



## You operate

- ✓ The worker nodes, the data plane your pods run on.
- ✓ Your workloads and how they are deployed and scaled.
- ✓ Your VPC networking and the cluster access control.

# Conformant Kubernetes, integrated with AWS

The Kubernetes is certified-standard, so your skills and manifests port anywhere. At the same time it is wired into the AWS services you already use.

## Upstream-conformant

- ✓ Certified standard Kubernetes, not a fork or a look-alike.
- ✓ Skills, tools, and manifests transfer to any conformant cluster.
- ✓ No lock-in to a vendor dialect.

### RUNS ANYWHERE

kubectl

Helm

YAML manifests

CRDs

## AWS-integrated

- ✓ Cluster and API access run on your existing AWS IAM identities.
- ✓ The cluster runs inside your VPC, on your subnets.
- ✓ Load balancers, EBS storage, and ECR registries are native services.

### NATIVE ON AWS

